

PART ONE

1.

For the purposes of this homework, my user population is, broadly, people who want to read and watch books, movies, and TV shows. The population is likely high-school-age and above, especially middle-aged and older adult populations who may have more interest and fewer resources. The population is not especially tech-literate and are not familiar with existing resources.

This user population is broad, but it is different from me because it trends older and is less familiar with existing resources.

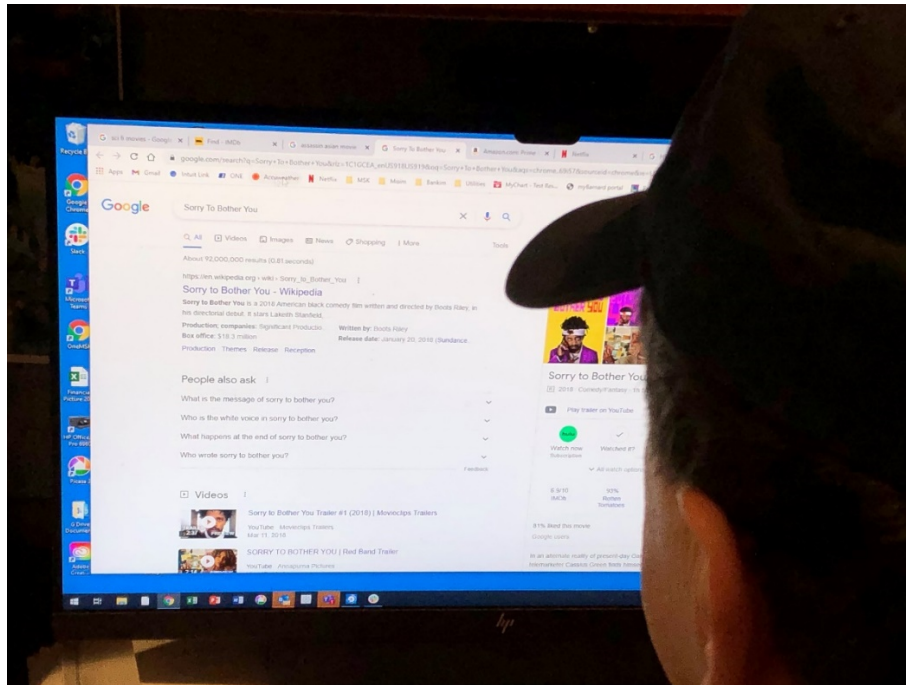
2.

The problem I think my userbase has is the lack of a convenient, organized way to find, collect, and access media content. Their friends recommend them content, but they cannot conveniently find and access it, so they forget it until they happen to find it again. They want to keep track of the media that they've heard of, whether it's been recommended to them or it's something they remember enjoying, but they find it difficult because it is held in a number of different and confusing sources, especially online sources (e.g., some are at a local library, some are on Netflix now but won't be in a month, some are free online but only if you know to check a certain website, etc.)

PART TWO

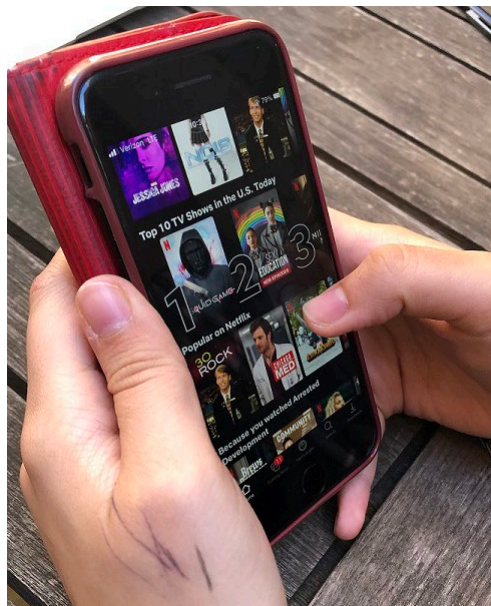
1.

USER 1:



This image shows User 1 looking at an open browser window. The window has a number of media-finding tabs open, including IMDB, Google searches, and streaming services. The screen he is looking at—a Google search for the movie ‘Sorry to Bother You’—only shows one of the streaming services that the movie is available on.

USER 2:



This images shows User 2 browsing through Netflix on her phone. She prefers to browse casually, so she is looking at the movies and shows which are most popular on Netflix currently.

2.

USER 1:

User 1 is an Israeli man in his early sixties who works in the technology industry. He watches new media with his wife very often. He enjoys searching for new media, but he is dissatisfied with his current process and thinks it could be made more efficient.

During my observation, he walked me through how he searches for new movies and shows. His primary method is to open several websites at the same time, usually IMDB and his favorite streaming services, like Amazon Prime, Netflix, and sometimes Hulu. These streaming services algorithmically recommend movies and shows. When he finds one that appears unfamiliar, he searches the title in IMDB to find a summary and trailer. If he finds it interesting, he returns to the streaming service and adds the movie or show to his watchlist.

His secondary method is to look for a specific genre of movie or show, such as sci-fi. In this method, he searches Google for movies of that genre, opens a site that he trusts, and looks through lists of recommended movies. He again searches unfamiliar titles in IMDB. However, these online lists don't tell you how the movie is available, so he also searches Google to locate the movie, and only then goes to that service to add the movie to his watchlist. When he demonstrated, Google incorrectly showed only one of the streaming services the movie was available on (one he doesn't use), so he had to search again.

Finally, if he knows what he's looking for, he uses Google to check what streaming services have that movie or show. However, he finds this unreliable because many movies have similar names, and sometimes he doesn't remember the title fully or accurately. When he remembers to do so, he also checks his library website. (When he demonstrated, the library website was slow to search, and did not have the particular title he wanted.) However, he usually gives up the search before he remembers to try the library site. He also has difficulties using Hoopla. Hoopla provides streaming services for a laptop, and he primarily uses a television. He finds it inconvenient to connect a laptop to his television in order to watch movies with his wife.

In the interview, he stated that he far prefers having friends recommended media to any of his other methods. When this occurs, he writes and sends a note to himself, and searches for the media through the above methods. Above all, he prioritizes finding media through reviews that he trusts—which is to say, from people with similar tastes to him, usually friends or family.

Finally, he added that finding books is an issue that bothers him tremendously. He has exhausted the books by authors he likes, and it is difficult for him to find new books which he thinks he will enjoy. The traditional book review sites that he uses, like the New York Times, rarely review the genres he prefers (science fiction, fantasy, mystery.) Searching Google for recommendation lists, like he does for movies, usually gives him lists of older books which he has already read or doesn't want to read.

This interview was highly informative. I learned that there is, indeed, room for improvement: User 1's media-finding process was in many ways inefficient, and broke down at a number of stages. I saw a user need for *concentrated information*, so that instead of going to many sources, User 1 could go to one source. I also learned that there exists a user need for *trusted recommendations*, either from friends or family or from algorithms like those on streaming sites.

USER 2:

User 2 is a Barnard student majoring in physics. She occasionally looks for new media, but uses it as a means to an end rather than enjoying the process. She is also likely to rewatch shows she enjoys. She is satisfied with her current media-finding and media-watching experiences.

During my observation, User 2 demonstrated how she finds new movies and shows.

When finding new television shows, she prefers to browse through the Netflix homepage on her phone, especially the shows which are currently popular on Netflix. She prefers shows which other people have recommended her or which she has heard of before, because it gives her something to talk about with others. However, she usually does not look for anything in particular, preferring to let herself be drawn to these shows naturally. She does have lists of recommended shows, but she doesn't reference them. When she wants a particular genre, she goes to Google and looks through Google's recommended list until she finds a synopsis that is engaging, then looks it up on Google again to find which services host it. Aside from browsing, she also looks through the 'currently watching' portion of Netflix to see if any of her current shows are engaging for her.

When watching movies, she references her to-watch list in the Notes App on her phone and sees what she is in the mood for. She then searches for the movies on Google.

Her main issue with finding media is the Google search. Google is occasionally wrong about which streaming services the media is available on. She also uses a number of other sites to acquire more information, like Rotten Tomatoes to find reviews or Does the Dog Die to find content warnings.

In terms of finding books, she has a list of books to read which she keeps in mind. However, she usually reads 'simpler' books by finding free PDFs online. She uses Google to find books by looking up authors and utilizing Google's 'books like this' section to find books she hasn't read. She then looks up the ratings and (sometimes) a synopsis, usually landing at Goodreads. During the contextual inquiry, she found that the book she was looking for was 'highly emotional,' and decided not to read it, since she wanted something lighthearted. She also finds books through fanfiction, because fanfiction allows her to filter based on the literary tropes she enjoys.

While walking me through her process, User 2 told me a number of her likes and dislikes. She liked the Instagram-story-like mini-trailers which Netflix Mobile used to have. She likes algorithmic recommendations more than recommendations from friends, and if she cannot easily find a media title that was recommended to her, she usually gives up. She would prefer not to have to create a new account on a new platform. She prefers watching shows which are new and broadly popular, but would enjoy having multiple streams of recommendations based on her differing wants. Finally, she engages with media content almost exclusively through her mobile device.

This interview also taught me a lot. User 2's needs are, above all, for *convenience*. She likes having media content immediately available to her, and though she usually browses through the entire Netflix homepage, she dislikes "digging" for specific content. However, like User 1, she also has a need for concentrated information: her process for finding movies is highly involved, leading her to go to a number of different sites, which may contribute to her preference for easily-accessible television shows.

3.

USER 1:

User 1 had numerous breakdowns in his process. Above all, his process was in itself a workaround: he didn't have a single source collecting movie and show titles, so he had to separately find them, research them, and find a way to access them. His process for finding books, more interestingly, was nonexistent: he simply can't find new books to read.

When demonstrating to me how he searches for a particular movie on Google, he wasn't able to find it for several minutes, even with the accurate title, until he narrowed down the search terms. Also, when showing me how he finds media content on his library website, the site was slow to search, and did not have the particular title he was using as an example. Both these breakdowns suggest to me that User 1's method, while more-or-less functional, was neither fully effective nor entirely efficient.

USER 2:

User 2's media-finding process seemed well-honed, but it also had a number of breakdowns. It took her some time to find her watching history in Netflix's mobile format. She also had to go to several sites in order to find a book she wanted to read: she first searched for the author on Google, then located a new book she hadn't read, then found reviews, and only then saw that the book was not something she would want to read.

4.

My target population has not changed, but the emphasis shifted slightly. Firstly, my target population can be expanded to the tech-literate. Tech-literate users also have issues locating and accessing new media. Secondly, while I can still design for a younger adult population, these users may already have honed processes for finding media, and may not want to shift. They may also have less desire for recommendations from friends, finding existing algorithmic recommendations sufficient for their needs. Older adult populations, on the other hand, appear more likely to have a need for new media-finding and -recommending methods.

5.

My problem statement still more-or-less applies given the results of my contextual inquiry, but has some important additions.

Information about the availability of media is not the end-all issue that my users face. While it is still an issue—both users cited Google's inaccurate information about what streaming services media is available on as a problem—users also want good recommendations for new media, and a source to find more information about particular titles, especially synopses, trailers, content warnings (or favored tropes), and how popular or current the media is.

Also, my users emphasized that they want media content tailored to their desires. I had assumed that people wanted to watch what their friends were watching, but this is not fully accurate. It is still a partial consideration (e.g., User 2 wanting to watch popular shows so she could talk about them with her friends), but it is not a primary need. Instead, they prefer to watch shows which they believe they themselves will enjoy.

6.

I think my userbase wants, but does not have, a concentrated source of recommendations and information about media. They want to find availability, popularity, individually-applicable reviews, trailers, summaries, and content warnings. They also want to be able to see what content would be most enjoyable to them based on what people with similar tastes have enjoyed. Their problem is that they do not have an efficient or convenient way of finding this information.

PART THREE

1.

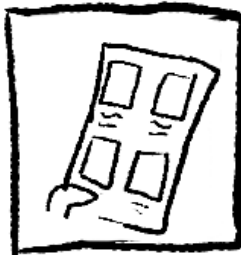
Steven Sokol is a 34-year-old lower-level manager who works in a large banking firm. Steven is a big fan of detective shows. There's nothing he likes better than to relax at the end of a long day at work with a good armchair mystery on his television, as long as it's not too violent. He prefers streaming, but because he likes to watch one show at a time, he doesn't subscribe to many streaming services at once. However, he's almost reached the end of his current show, *Midsomer Murders*, and he's looking for a new one. His friends at work have told him about the shows that they like, but he can never remember to write them down. He's already tried a few new shows on his favored streaming service, but none of them are the kind he likes—one was so violent that he couldn't watch past the first ten minutes! He's worried that he won't be able to find a new show—one that he has access to and knows he will like—to unwind with in the evenings.

2.

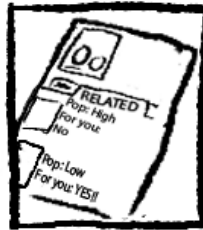
FIRST DESIGN:



On the subway back from work, Steven Sokol is looking for a show to watch—one he'll like, that's on a streaming service he has.



His app shows him some shows he might like, but he wants something as similar as possible to his current one...



...so he looks it up + finds similar shows, with a 'rec' value showing how much he'll like them! He clicks on a show.



There is also a summary, content warnings, and reviews. If he wants, he can see how similar the reviewer's tastes are to his own. He adds it to his watchlist.



Steven puts his phone down. Now, when he gets home, he knows what he's going to watch!

SECOND DESIGN:



Mei is bored and wants to see what's available to her without clicking around too much.



The website shows her all the media on the different platforms she has access to—plus what her friends have recommended her.



When she clicks on a title, she can immediately find other sites with info about it, her friends' recs, + a way to watch it ASAP.



She clicks a button to get redirected to Netflix. She's ready for a fun evening in!



And afterwards, she can tell her friend what she thought of it—or recommend it to someone else!

3.

The first design allows users to find media content based on similarities to existing media or based on recommendations (such as an algorithm based on the user's watch history and the ratings they

have given.) Once a media title is found, the user can find information on it, such as reviews, content warnings, and what streaming platforms it is available on. The user can add it to their watchlist on the site, or, if they have access to one of the relevant platforms, open a new tab directly in the platform and start watching it. The emphasis is on *recommendations and information*.

The second design shows users the media they have access to based on the hosting platforms that they are available on. It allows the hosting platforms to use their own recommendation algorithms, while adding the functionality of other, ‘helper’ websites, like Goodreads, Rotten Tomatoes, and IMDB for reviews or Does the Dog Die for content warnings. There is also a strong social element: you can recommend media to your friends and have them recommend media to you. The emphasis is on *concentration of disparate sources*.

I believe the first design has more promise. It takes the user directly to the media, without the interference of the hosting platforms. This also means that the user can find all the platforms that a given title is available on (e.g., a movie may be available for money on Amazon Prime, but free on Netflix, if you have Netflix, or available online through your local library.) I believe the ideal user experience is to go directly from searching to watching or reading the media content.

4.

FIRST DESIGN:

Strengths:

- (1) Focus on the media. The user doesn’t want to see the different platforms that they own; they want to see the media that they want.
- (2) Holistic recommendations, concentrated information. Users can get recommendations for shows on Hulu based on their Netflix watch history, or find that a movie they want is free from their library before they pay to watch it on Amazon Prime.

Weaknesses:

- (1) Repeating work. Many platforms have their own, highly-tuned recommendation algorithms; it would be difficult to generate an algorithm that could compete.
- (2) Cost and time. Due to the above reason of repeating work, this design would be difficult and costly to achieve.

SECOND DESIGN:

Strengths:

- (1) Show the user what they’re paying for. The user knows they have access to a number of streaming platforms; by showing how the website is concentrating all the information in one place, they can see how they are given convenient access to the platforms they enjoy.
- (2) Transparency. The user knows exactly what’s going on: the information from different platforms is being concentrated into one easy-to-access hub. There is no black box of site recommendations, nor privacy concerns.

Weaknesses:

- (1) Broken user experience. Dividing by platform leads to the same media shown in different places, and takes the user out of the experience of finding and watching shows.
- (2) Hiding content. If a user doesn’t have access to a particular title, the site will not show it—even if it’s something the user would like so much that they would pay for access!

5.

Through my contextual inquiry, I found that users dislike having to switch between different platforms, and enjoy accurate algorithmic recommendations rather than specifically wanting recommendations

from their friends, as I had expected. My most promising design therefore focuses on maximizing the effect and accuracy of the algorithmic recommendations, and minimizing the experience of going between different sites. I included a recommendation for popular shows in my storyboard, reflecting User 2's desire to watch popular shows.

PART FOUR:

1.

My problem statement is challenging and ambitious because of its large scope and technical and practical difficulties.

One of my project's foremost difficulties will be maintaining accurate, up-to-date information about media platforms. The purpose of the project is to concentrate accurate information and access in one space. Both of my users relied heavily on Google's options for searching for media, and found its inaccuracies to be a problem. In order to be at all attractive, my project needs to provide a UI that is more reliable and convenient than Google searches—which is a tall order. The project is also technically difficult, due to the large amounts of data required, the need for collaboration with a number of media-providing platforms, and the need for refined and individually-relevant recommendations.

2.

Right now, existing design solutions for gaining information and access to new media are decentralized. Sites with media reviews, like Goodreads or Rotten Tomatoes, are separate from sites with information, like IMDB, or sites with access to media, like Netflix or Hulu. Sites with recommendations are either media-providing sites, whose recommendations are only based on the media on their specific platform, or Google, which has limited personalization. The most viable option for finding where media can be accessed is Google, which is also separate from the other sites, and is occasionally inaccurate.

Existing user strategies involve either switching between all these different solutions, which is inefficient, or by limiting the breadth and depth of their searches, which causes users to find subpar media. My design is original because it concentrates these user needs in one place.

3.

In order to carry out the Individual Project successfully, I will need to account for a number of risks.

Design risks include *convenience*. I will need to pick and choose what functionality to include in my project. There is a lot that can be added, but if I put in too much, my solution may become inconvenient to use—which would be a big problem, since convenience is my project's main draw.

Another one is *targeting my audience*. Am I looking for people who want a concentrated hub that is specifically an intersection of books, movies, and TV? Or do my users want a concentrated hub to get more information about movies alone, or movies and TV and not books, or books alone? I will need to find out who exactly my project is for, or else my project may be misaimed.

There are also some major technical challenges. The project needs to be able to identify a large number of media titles, and provide accurate and up-to-date information about each one, as well as functionality for redirecting the user to specific streaming platforms or adding shows to that platform's watchlist. Further, it would ideally have a trustworthy recommendation algorithm.

4.

I plan to mitigate the risk of overcrowding the project's design by identifying user priorities. If users prioritize one aspect of functionality over another (e.g., they may be very interested in reviews, but not care about finding content warnings), I can emphasize the more important functionality, keeping the design clean and convenient for the user.

In order to ensure that I am targeting the right audience, I will do something similar: I will use interviews, prototyping, and evaluation to see what users find most interesting about my project, and what functionality is most important to them.

The technical challenges are less relevant in terms of this project, since I will not be implementing it. The recommendation system can use collaborative filtering—finding users with similar tastes by determining which users have given similar ratings to the same media, and finding other media content that those users enjoyed. Further, it is possible that some challenges could be alleviated by integrating already-existing database sources into the site. For example, the site could use the functionality of IMDB rather than have its own database of information about movies.

5.

I am most looking for feedback on the risks involved and the challenge of the project, and where to go from here. I'm not certain about how to judge the risks or the challenge, and would appreciate help with this. I'd also like some feedback about the central conceit of the project: is this a good idea in the first place? Where can I go with it in future iterations?